Shock Absorber Design and Multiphysics Simulation Report

Software Used: Autodesk Inventor Professional & ANSYS Mechanical APDL
Project Type: Individual Academic/Engineering Simulation Project
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1. Introduction

This project focuses on the **design and simulation of a mechanical shock absorber**, simulating real-world behaviour under various mechanical and vibrational conditions. The workflow includes 3D modelling in Autodesk Inventor, followed by static, modal, and random vibration analysis in ANSYS Mechanical APDL.

2. Objectives

- To design a functional shock absorber assembly with mechanical adjustability.
- To analyse its structural behaviour under static compression.
- To extract modal frequencies and ensure the system avoids resonance.
- To evaluate response under realistic random vibration using PSD input.

3. CAD Modelling – Autodesk Inventor

Components Created:

- Main Rod / Body
- Plunger / Base
- Threaded Nut (adjusts spring tension)
- Helical Spring / Coil

Each component was created as a separate part file and assembled using fully constrained joints.

4. Material Assignment

Component	**Material**
Spring/Coil	Structural Steel

Component	Material
All Other Components	Gray Cast Iron

5. Import and Simulation – ANSYS Mechanical APDL

Simulation Types:

- Static Structural Analysis
- Modal Analysis
- Random Vibration Analysis (Base Excitation)

Meshing:

- Mesh Element Size: **0.003 m**
- Finer mesh applied around contacts and stress concentration zones

6. Contact Definitions

Contact Pair	Type	Coefficient of Friction (μ)
Rod ↔ Base	Frictional	0.1
Spring ↔ Base / Nut	Frictional	0.3
Spring ↔ Rod, Rod ↔ Nut	Bonded	—

7. Boundary Conditions and Loads

- **Displacement:** -10 mm applied in Y-axis to rod
- **Support:** Fixed support at plunger/base

8. Static Structural Analysis

A. Total Deformation

Time [s]	Max [m]	Avg [m]
0.2	2.02E-03	1.62E-03
0.4	4.04E-03	3.23E-03
0.7	7.07E-03	5.65E-03
1.0	1.01E-02	8.08E-03

B. Equivalent Elastic Strain

Time [s]	Max [m/m]	Avg [m/m]
0.2	2.93E-03	1.04E-04
0.4	5.80E-03	2.05E-04
0.7	1.01E-02	3.57E-04
1.0	1.44E-02	5.09E-04

C. Equivalent (Von Mises) Stress

Time [s]	Max [Pa]	Avg [Pa]
0.2	4.28E+08	1.74E+07
0.4	8.54E+08	3.45E+07
0.7	1.49E+09	6.02E+07
1.0	2.13E+09	8.58E+07

9. Modal Analysis

Mode Frequencies:

Mode	Frequency [Hz]
1	657.5
2	697.47
3	853.57
4	980.78
5	1152.4
6	1197.8

10. Random Vibration Analysis

Power Spectral Density (PSD) Input:

Frequency [Hz]	Acceleration [$(m/s^2)^2/Hz$]
5	100
10	150
20	200
30	200
50	100

- PSD-based vibration loading applied along base in vertical direction.
- Results include peak stress, deformation, and vibrational response at resonant frequencies.

11. Project Outcomes

- Spring component showed the **maximum deformation** under compression.
- Highest stress regions were observed at **contact interfaces**, especially between nut and coil.
- Modal analysis confirmed **safe frequency separation**, reducing the risk of resonance.
- Random vibration results showed the system remained within material limits under dynamic loading.

12. Key Learnings

- Gained hands-on experience with **CAD-to-FEA simulation workflow**.
- Learned to implement **realistic frictional and bonded contacts** in simulation.
- Understood how to interpret and validate results for both **static and dynamic behaviour**.
- Developed a stronger grasp on **modal and random vibration analysis techniques**.